Anglo-Chinese Junior College

JC2 Biology Preliminary Examination
Higher 2

A Methodist Institution
(Founded 1886)

CANDIDATE NAME		FORM CLASS	
TUTORIAL CLASS		INDEX NUMBER	

BIOLOGY **9744/03**

Paper 3 Long Structured and Free-response Questions

30 August 2022
2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing paper(s)

READ THESE INSTRUCTIONS FIRST

Write your Name, Class and Index number in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer any **one** question on the separate writing paper(s) provided.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all the writing paper(s) used securely together.

For Examiners' use only	
Section A	
1	/ 30
2	/ 10
3	/ 10
Section B	
4 or 5	/ 25
Total	/ 75

For Examiner's Use

Section A

Answer **all** the questions in this section.

1 The Human Immunodeficiency Virus (HIV) is an enveloped virus that consists of glycoproteins on its surfaces. Fig. 1.1 shows the structure of an Env glycoprotein on HIV.

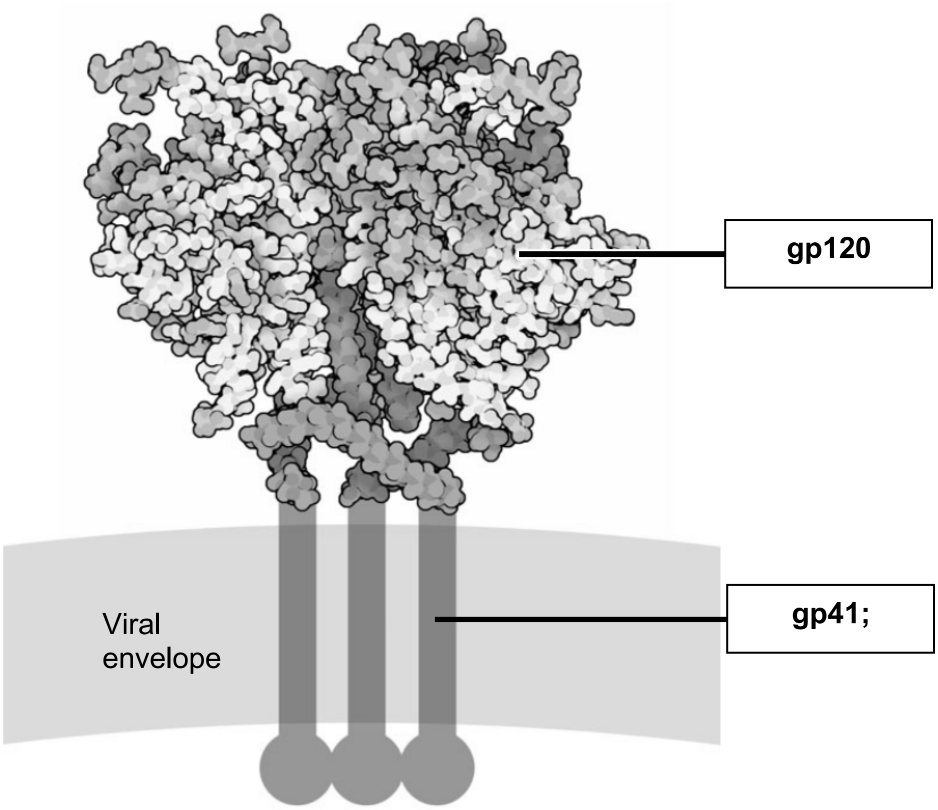

Fig. 1.1

(a) **(i)** The Env glycoprotein is made up of two associated subunits.

In the boxes in Fig. 1.1, state the name of these two subunits.

@1m for correctly labelled gp120 and gp41 [1]

(ii) Describe how the glycoprotein interacts with the viral envelope.

1. **The glycoprotein contains amino acid residues with hydrophobic/ non-polar R groups which can interact with the hydrophobic hydrocarbon tails/ fatty acid chains of the viral envelope;**

2. **Forming weak hydrophobic interactions;**

@1m each [2]

For Examiner's Use

(iii) HIV envelope proteins are glycosylated by host cell enzymes as HIV genome does not encode gene products capable of synthesising carbohydrates.

Suggest how glycosylation of HIV envelope proteins may impair the host immune response to HIV infection.

1. **The carbohydrates are added by host enzymes thus are 'self' glycans/ disguised HIV envelope proteins as 'self'/ self-antigens/ not foreign;**
   OR
   carbohydrates added are able to shield HIV surface proteins/ result in a change in the conformation of glycoproteins;

   ....

2. **prevent recognition/ binding by the immune cells or antibodies/ evades detection by immune system;**

....

@1m each

.... [2]

(b) Describe the interactions between HIV and the host cell to enable the entry of the virus into the cell.

1. **gp120 on the envelope of HIV binds to the CD4 receptor and a co-receptor on the helper T cell/ host cell;**

   ....

2. **This binding triggers an allosteric/ conformation change in gp41, which penetrates the host cell surface membrane;**

   ....

3. **The envelope membrane of the virus fuses with the host cell surface membrane;**

   ....

4. **The nucleocapsid is released into the cytoplasm, the capsid is degraded and the HIV genome and enzymes are released;**

....

@1m each, max 3m

.... [3]

[Turn over

Upon entry into the host cell, HIV uses the enzyme reverse transcriptase to reverse transcribe its RNA genome into double-stranded DNA for integration into the host genome.

Reverse transcriptase is made up of two subunits, forming two different active sites that are essential for its function. Fig. 1.2 shows the location of these active sites on the enzyme with its newly formed double-stranded DNA.

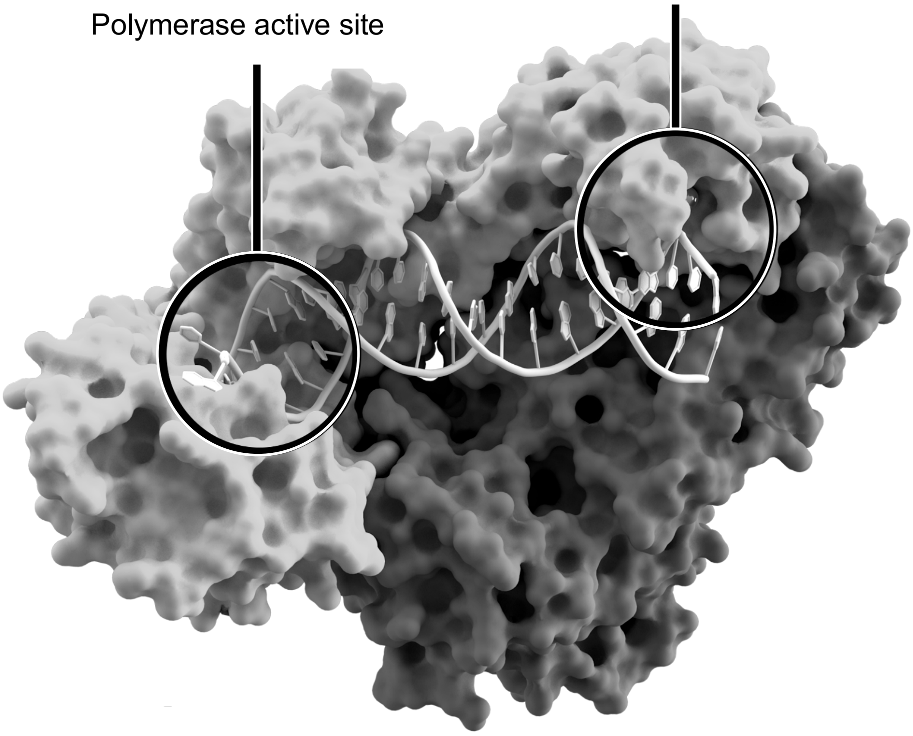

Fig. 1.2

(c) **(i)** Explain what determines the precise positions of these two active sites in the structure of reverse transcriptase.

1. **Primary structure consisting of unique/specific number and sequence of amino acids linked by peptide bonds;**

....

2. **folded into α-helices and β-pleated sheets that make up secondary structure maintained by hydrogen bonds between the –CO group and the –NH group of the polypeptide backbone;**

....

3. **Further folding results in 3D conformation/ globular protein/ tertiary structure maintained by hydrogen bonds, ionic bonds, disulfide bonds, and hydrophobic interactions (any 2 bonds) between the R groups of amino acids, forming active sites;**

....

4. **Association of two subunits by hydrogen bonds, ionic bonds, disulfide bonds, and hydrophobic interactions (any 2 bonds) between the R groups of amino acids in the quaternary structure, forming active sites;**

....

For Examiner's Use

@1m each

.... [4]

(ii) Compare the structures of reverse transcriptase and DNA.

Similarity:

1. **Both are (two) polymers formed by condensation reactions;**
2. **Both are held by hydrogen bonds / hydrophobic interactions;**

....

Differences:

2. **Reverse transcriptase is made up of amino acids while DNA is made up of deoxyribonucleotides;**
3. **Reverse transcriptase can be made up of 20 common types of amino acids while DNA can be made up of 4 types of nitrogenous bases;**
4. **Monomers in reverse transcriptase are held by peptide bonds while monomers in DNA are held by phosphodiester bonds;**
5. **Polypeptides in reverse transcriptase are held by ionic bonds, hydrogen bonds, hydrophobic interactions while polynucleotides in DNA are only held by hydrogen bonds, hydrophobic interactions;**
6. **Reverse transcriptase has a globular structure while DNA is helical;**

....

....

@1m each, 1m for similarity and 1m for difference

.... [2]

[Turn over

One application of reverse transcriptase in research is in a technique called reverse transcription-polymerase chain reaction (RT-PCR). The classical PCR technique can only be applied to DNA strands, whereas RT-PCR allows the application of PCR technique to an RNA template. A potential use of RT-PCR is in the study of gene expression.

Fig. 1.3 illustrates the process of RT-PCR using mRNA as a template. Oligo dT primers, which are single-stranded sequences of repeating deoxythymine, are added to the reaction mixture.

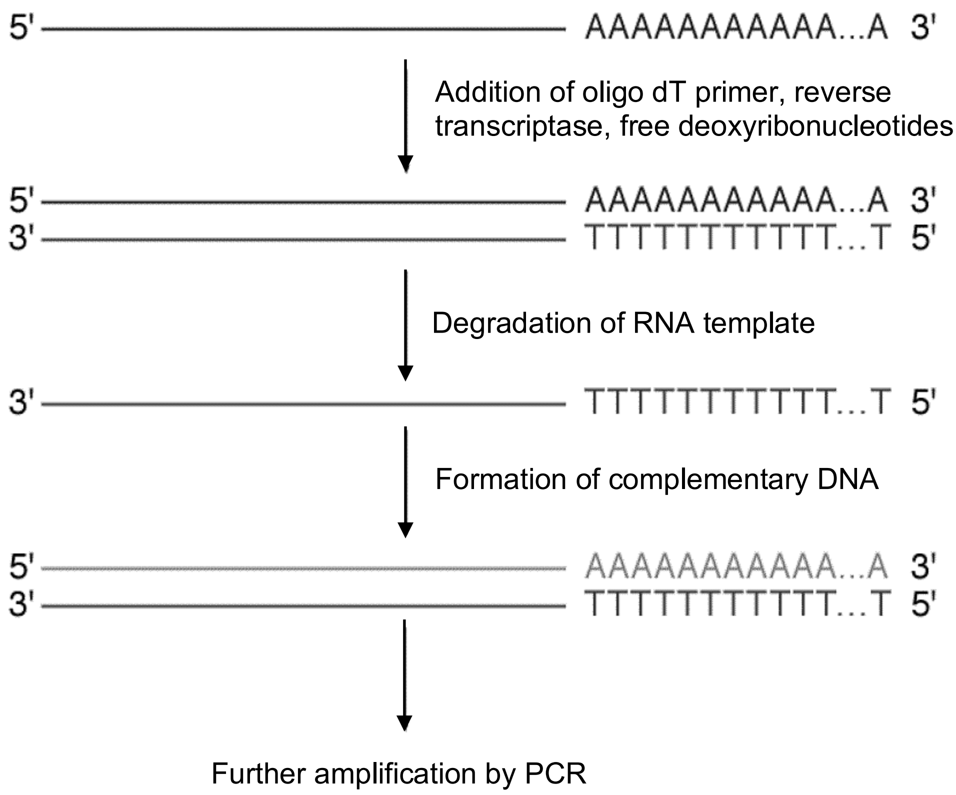

Fig. 1.3

(d) **(i)** With reference to Fig. 1.2 and Fig. 1.3, explain how reverse transcriptase and oligo dT primers can be used in RT-PCR to study gene expression.

1. ***Oligo dT primers bind to <u>poly-A tail</u> of mRNA via <u>complementary</u> base pairing;**
2. ***Presence of mRNA / formation of cDNA / amplification will indicate that the gene has been transcribed;**

....

3. **Reverse transcriptase catalyse the synthesis of a DNA strand complementary to mRNA template at its <u>polymerase active site</u>;**

....

4. **Hydrolysis/ degradation of the RNA strand the occurs at its <u>nuclease active site</u>;**

....

5. **Reverse transcriptase then catalyses synthesis of a second DNA strand complementary to the first at its <u>polymerase active site</u>;**

For Examiner's Use

@1m each, mp1 or 2 compulsory, max 3m

.... [3]

The relative activity of reverse transcriptase at different temperatures can be investigated using the SYPRO orange fluorescence assay. The reverse transcriptase is incubated with a fluorescent chemical known as SYPRO orange and the temperature is gradually increased. SYPRO orange interacts with the hydrophobic regions in an enzyme which are normally not exposed. The amount of SYPRO orange which binds to an enzyme can be detected by measuring the intensity of the fluorescence.

Fig. 1.4 shows the results of the SYPRO orange fluorescence assay on two types of reverse transcriptase.

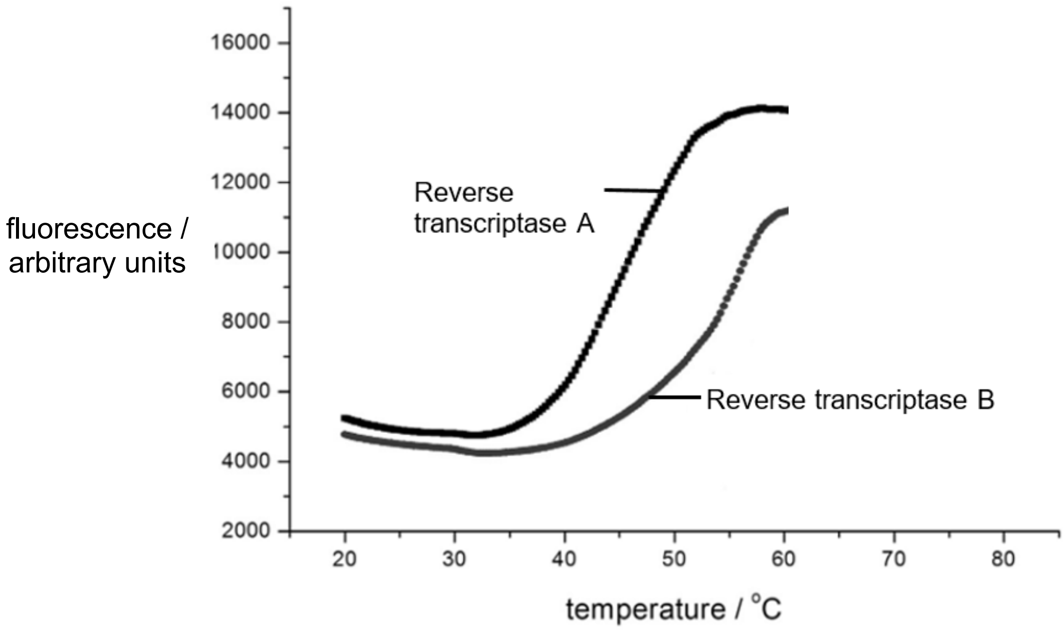

Fig. 1.4

(ii) During RT-PCR, a high incubation temperature may be used.

Suggest and explain which reverse transcriptase is more suitable to be used for RT-PCR.

1. Reverse transcriptase B;

....

2. As temperature increases, the enzymes denatures/ unfolds, exposing the hydrophobic regions for the binding of SYPRO orange, resulting in an increase in fluorescence intensity;

....

3. At all temperatures from 20 to 60°C, fluorescence level of reverse transcriptase B is lower than that of reverse transcriptase A;

....

4. which shows that reverse transcriptase B is more thermostable/ more heatstable/ withstand high temperature/ denature to smaller extent/ denature at higher temperature and likely to remain active during the reaction;

[Turn over

For Examiner's Use

@1m each

.... [4]

Infections by some viruses are known to increase the risk of cancer development. The HIV weakens the immune system and increases the susceptibility of the body to infections which cause cancer. Other viruses like the human papillomavirus (HPV) can integrate its genome into the host genome and produce highly carcinogenic proteins.

Fig. 1.5 shows how a HPV infection which persists for many years can lead to cellular changes that, if untreated, may result in the formation of a malignant tumour.

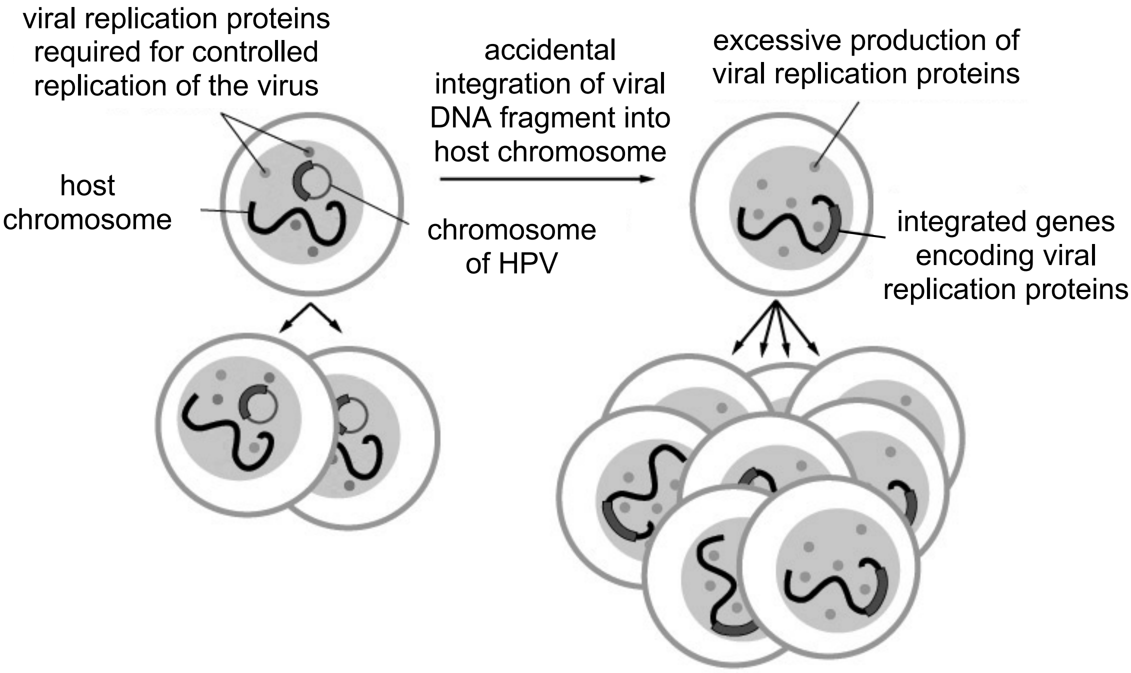

Fig. 1.5

Fig. 1.6(a) shows how proteins regulating the cell cycle checkpoints, such as p53 and Rb protein, keep the cell cycle in check and prevent cell proliferation in a normal cell. Fig. 1.6(b) shows an example of two viral replication proteins, E6 and E7, interfering with p53 and Rb protein function and resulting in excessive cell proliferation.

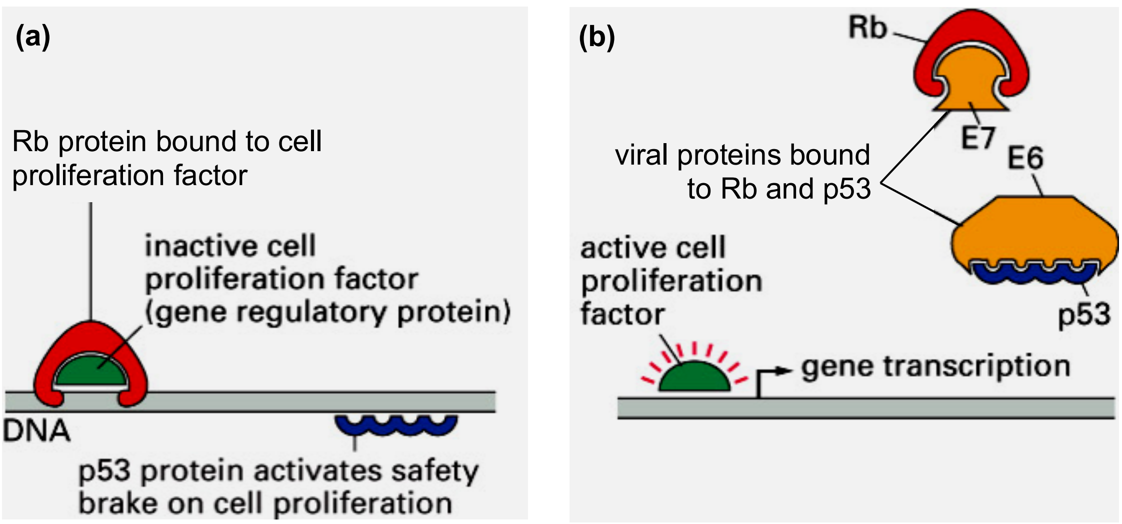

Fig. 1.6

For Examiner's Use

(e) **(i)** With reference to Fig. 1.5 and Fig 1.6, suggest and explain how a tumour can be formed in patients with HPV infection persisting for many years.

1. **Integration of viral DNA fragment containing genes coding for viral replication proteins into the host cell's genome;**

2. **under the control of an active promoter resulting in excess production of viral replication proteins;**

3. **E6 binds complementary to and sequester p53, prevent cell cycle arrest;**

4. **E7 binds complementary to and sequester Rb protein, prevent inhibition/ results in activation of cell proliferation factor;**

5. **Decreases fidelity of cell cycle checkpoints such that cells with DNA damage can continue to divide uncontrollably;**

@1m each, max 4m

[4]

(ii) Suggest a reason why it may take 10 to 20 years for HPV-infected cells to become cancerous.

1. **HPV-infected cells may be detected and removed by immune cells / antibodies;**

2. **Time is required for accumulation of mutations in several tumour suppressor genes / at least one proto-oncogene / genes involved in carcinogenesis/angiogenesis/metastasis etc.;**

3. **Integration of HPV genome only occurs by chance/ is accidental (according to Fig. 15);**

4. **HPVs may remain in latency within the host cell and viral proteins are not produced;**

@1m each, max 1m

[1]

[Turn over

For Examiner's Use

HPVs spread easily through direct sexual contact, including vaginal, oral, and anal sex. Fig. 1.7 shows the HPV-associated cancer incidence in the United States between 2008 and 2012.

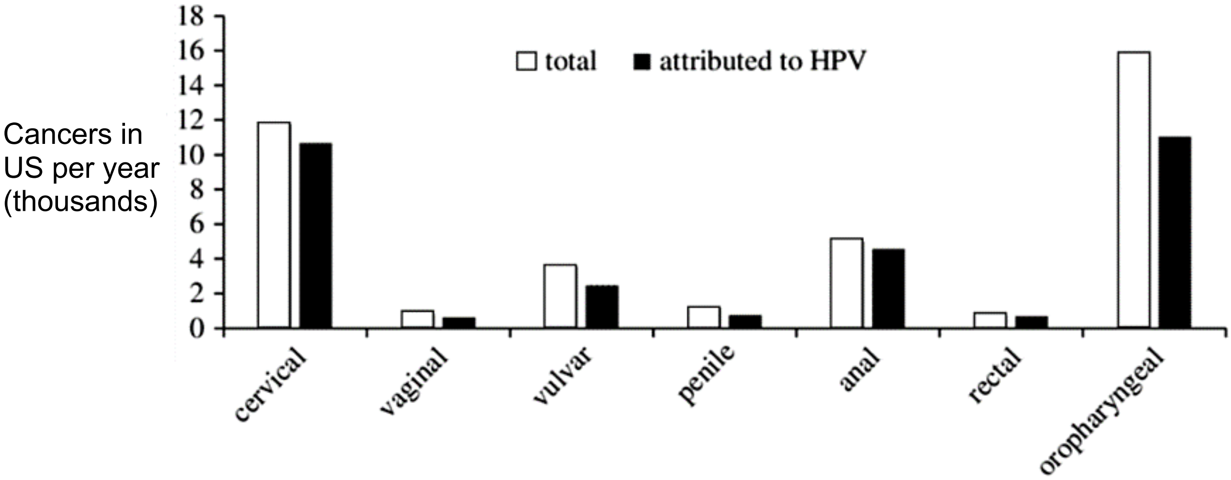

Fig. 1.7

The HPV vaccine was first introduced in 2006, with a recommendation to vaccinate females of ages 11 to 26. With growing evidence supporting the expansion of the vaccine, many countries are extending its coverage to males.

Fig. 1.8 shows the numbers of cancers caused by HPV and the gender distribution for each type of cancer in the United States each year.

Fig. 1.9 shows the estimated effect of HPV vaccination on oral HPV infections among individuals from 18 to 33 years of age in the United States population. Results are presented as the total number of infections in the absence of HPV vaccination, the number of preventable infections at 100% vaccination levels, and the number of potentially vaccine-prevented infections at current HPV vaccination levels in the population.

Fig. 1.10 shows the rates of HPV-associated cancers and age at diagnosis among men in the United States per year.

For Examiner's Use

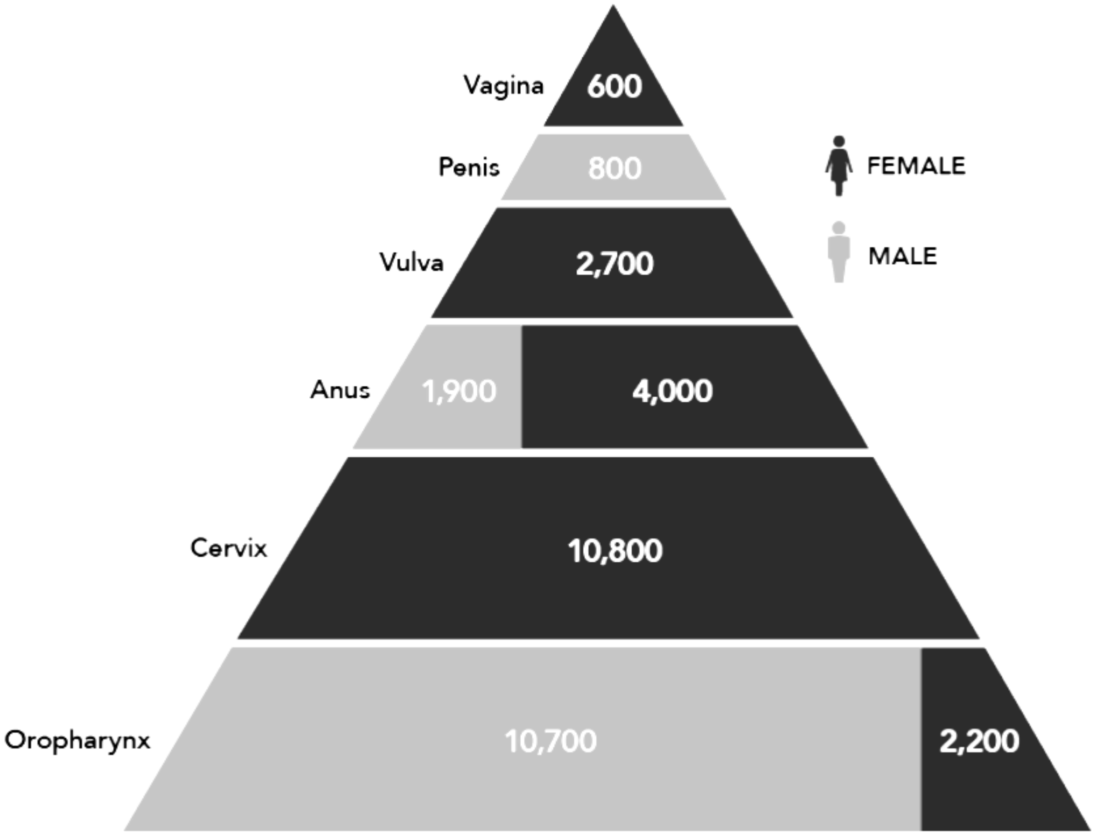

Fig. 1.8

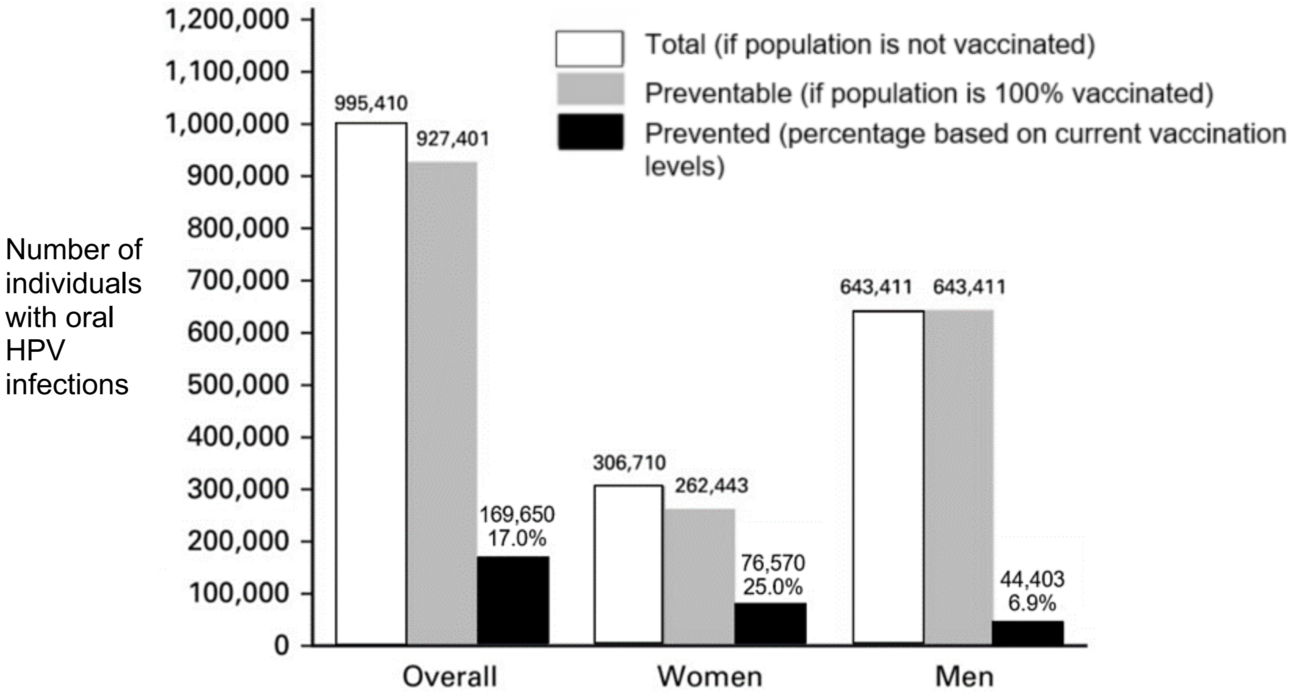

Fig. 1.9

[Turn over

For Examiner's Use

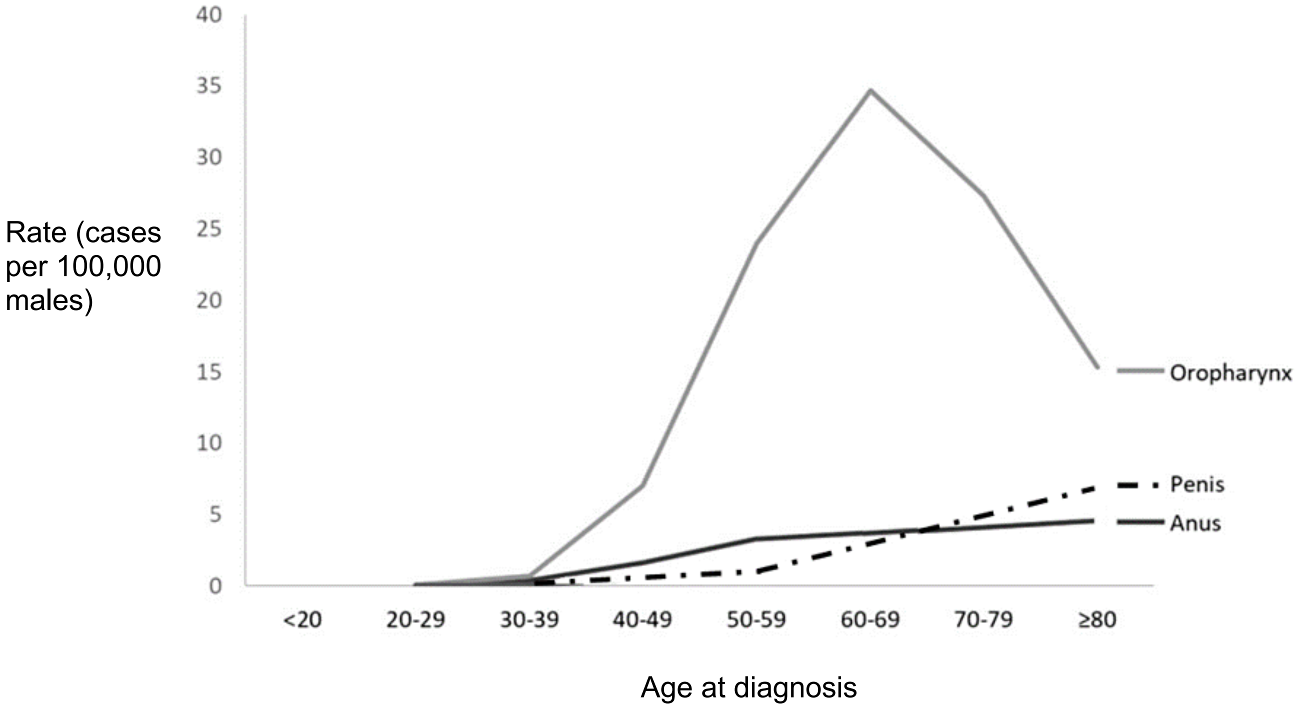

Fig. 1.10

(f) With reference to Fig. 1.7, Fig. 1.8, Fig. 1.9 and Fig. 1.10, discuss why HPV vaccination should be encouraged in young males less than 20 years old in the United States population to reduce the risk of HPV-associated oropharyngeal cancer.

1. **From Fig. 1.7, majority of oropharyngeal cancer, 11 thousands out of 16 thousands cancers in US per year, is attributed to HPV infection;**

2. **From Fig. 1.8, as many as 10700 cases of HPV-associated oropharyngeal cancer affect males, more so than females at 2200 cases;**

3. **From Fig. 1.9, number of infections prevented by current level of vaccination is low at 6.9%, which implies that current vaccination effort is insufficient;**

4. **From Fig. 1.9, vaccination can prevent 100% of oral HPV infections in men;**

5. **From Fig. 1.10, vaccination should be given earlier than 20 years old as a preventive strategy as the youngest males diagnosed with oropharyngeal cancer is between 20-29 years old/ males are beginning to be diagnosed with oropharyngeal cancer from 20 years old onwards;**

@1m each, max 4m

[4]

[Total: 30]

For Examiner's Use

2 The venom of a snake contains many protein toxins which can damage the tissues of a victim who has been bitten. The snake bite can lead to significant disability or death within hours, and an antivenom would be necessary for treatment. The following steps describe how an antivenom is traditionally produced.

- The venom of a snake is collected.
- An animal, often a horse, is injected with a controlled quantity of the venom.
- The horse's blood is withdrawn and the antibodies produced in response to the protein toxins are isolated.
- The isolated antibodies are purified and formulated as an injection.

The antivenom produced is effective only against the species of snake from which the venom is obtained.

(a) **(i)** State the type of immunity that is conferred by the antivenom when administered to the snake-bite victim.

Artificially-acquired, passive immunity;

.... [1]

(ii) Describe how antibodies in the antivenom may reduce the harmful effects of toxins in the snake venom.

1. **Through neutralisation where the binding of antibodies to the toxin prevents the toxin from binding to host cell receptors;**
2. **Through opsonisation, where the binding of the antibodies mark the toxin for phagocytosis by macrophages;**
3. **Through the activation of complement proteins, triggers the formation of pores in the cell surface membrane/ osmotic lysis of the targeted cell;**

....

max 1m

.... [1]

(iii) Explain why a particular antivenom is effective only against a specific species of snake.

1. **Venom from different species of snake contain different protein toxins;**

....

2. **During the production of the antivenom, only B lymphocytes with receptors that have a binding site with a 3D conformation complementary to the protein toxins undergo clonal selection;**
OR
The antibodies produced in response to the venom have antigen binding sites / variable domains that have a 3D conformation complementary to the protein toxins;

....

@1m each

[Turn over

For Examiner's Use

.... [2]

(iv) Suggest why injection with vaccines containing inactivated protein toxins is not an effective treatment for a snake-bite victim.

1. Vaccines requires several days to take effect, but a snake-bite victim requires immediate treatment;
OR
Vaccination is only effective if administered several days before the bite;

....

2. Time is required for antigen presentation/ activation of T and B lymphocytes/ differentiation of B lymphocytes into plasma cells/ synthesis or secretion of antibodies;

....

....

@1m each

.... [2]

Fig. 2.1 shows how the blood withdrawn from the horse is centrifuged to obtain three distinct layers – the blood plasma, a layer of white blood cells, and a layer of red blood cells.

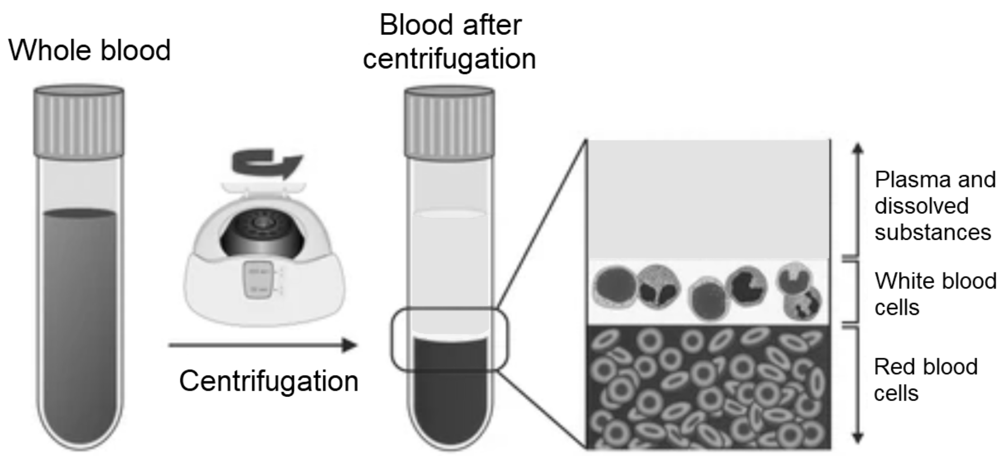

Fig. 2.1

(b) State the layer which the antivenom antibodies would be found in and explain your answer.

1. Blood plasma / Plasma and dissolved substances;

....

2. Antibodies are secreted/ released by plasma cells into the extracellular compartment, and are globular proteins hence soluble in water/ less dense than cells

@1m each

.... [2]

For Examiner's Use

Due to a global shortage in supply of antivenoms, scientists are exploring the use of genetically modified *Escherichia coli* to mass produce antivenom antibodies. The genes coding for the antibodies can be inserted into *E. coli* by transformation.

(c) Suggest one advantage and one challenge of using *E. coli* for antivenom production over the traditional method.

Advantage:

1. **More ethically acceptable as no animals have to be exposed to the venom;**
2. **Faster rate of antibody synthesis (due to faster rate of growth and cell division);**
3. **Less space required for production of antibodies/ more economical;**

…………………………………………………………………………………………………

Challenge:

4. **Method requires the sequence coding for the specific antibody to be known; OR**
5. **Efficiency of inserting the genes by transformation is low; OR**
6. **Bacteria cannot carry out post-transcriptional modification / excision of introns from mRNA; OR**
7. **Bacteria cannot carry out post-translational chemical modification required for antibody synthesis (due to lack of RER & GA);**

…………………………………………………………………………………………………

…………………………………………………………………………………………………

@1m max for advantage and @1m max for challenge, max 2m

…………………………………………………………………………………………… [2]

[Total: 10]

[Turn over

For Examiner's Use

3 Malaria is a mosquito-borne disease caused by a unicellular parasite, *Plasmodium*, which spends a part of its life cycle in a mosquito and a part of it in a human. The mosquito transmits the *Plasmodium* to a human when it feeds on human blood. In the low-lying coastal country of Belize, where malaria is a serious problem, studies have been made to determine the environmental factors which affect the incidence of the disease.

156 villages were studied over a ten-year period. Fig. 3.1 highlights the incidence of malaria in the different districts of Belize.

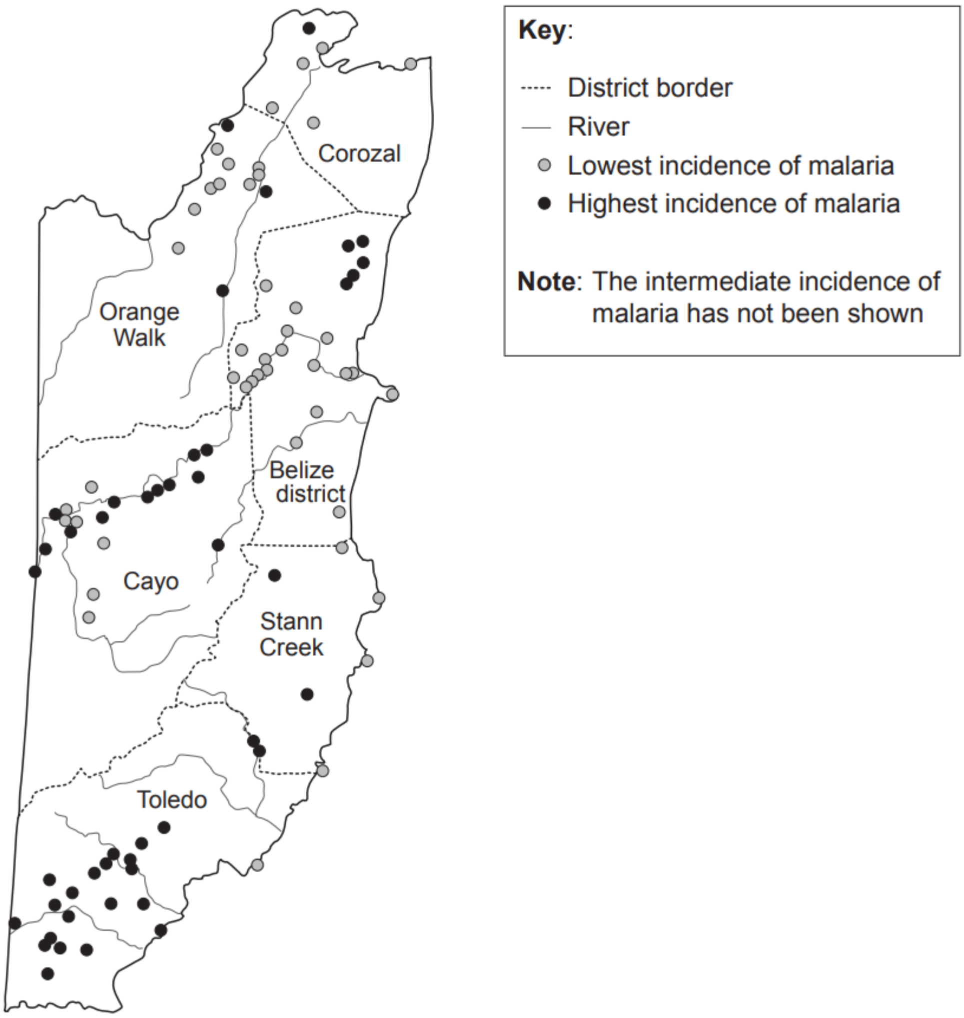

Fig. 3.1

(a) **(i)** Based on the studies conducted, explain why the association of rivers with a high incidence of malaria is inconclusive.

1. **Although high incidence of malaria can be found near rivers such as in Cayo/ Toldedo/ Orange Walk/ Stann Creek;**
2. **There are rivers with a low incidence of malaria along them such as the river in the Belize District;**
3. **There are rivers with a mixture of high and low incidence of malaria e.g. the river in Cayo;**

....

4. **Along one river in Toledo/Orange Walk, there is no incidence of malaria recorded at all;**

....

For Examiner's Use

5. **There is high incidence of malaria found away from rivers such as in the <u>Belize district/Toledo</u>;**

@1m each, max 2m

.... [2]

(ii) Suggest possible factors why the incidence of malaria is low in Corozal compared to other districts.

1. **It has less rainfall hence less stagnant pools of water present, preventing mosquitoes from breeding;**

....

2. **It experiences higher rainfall hence faster flowing rivers made conditions unfavourable for mosquito breeding;**

....

3. **More active predators present (such as amphibians) which prey on mosquitoes, reducing their numbers;**

....

4. **Inhabitants are more aware and practice methods (name/ describe a method) which keep the number of mosquitoes low;**

....

5. **The district has a smaller population hence the disease does not spread as fast;**
6. **AVP;**

@1m each, max 2m

.... [2]

Table 3.1 shows the total surface area for different types of land use in the country of Belize.

Table 3.1

Land Use		Thousands of hectare
Agricultural	Arable land (suitable for growing crops)	70
	Permanent crops	32
	Pastures	50
Non-agricultural	Forest area	1412
	Other lands	717

(b) **(i)** Calculate the percentage of land used for agricultural purposes in Belize.

1. **Total land used for agriculture = 70 + 32 + 50 = 152 thousands of hectare**
   Total land area in Belize = 70 + 32+ 50 + 1412 + 717 = 2281 thousands of hectare;

2. **152/2281 x 100% = 6.66% (3 sf);**

@1m each

[Turn over

For Examiner's Use

Percentage of land used for agriculture = **6.66** [2]

Climate change is affecting agricultural production and productivity in Belize. Mean annual temperatures have increased at an average rate of 0.1°C per decade since 1960 and climate projections suggest that temperatures could rise by another 1.8°C by 2050. Rainfall is likely to fall throughout the country, with decreases ranging from 7% in the northern zone to 10% in the southern zone. It has also been reported that some farmlands in the southern zone have a high incidence of malaria and pests.

(ii) Explain why converting some of the forests to farmland will worsen the effects of climate change.

1. **Burning trees will release carbon dioxide into the atmosphere;**
   OR
   Burning trees will remove carbon sinks as crops that are planted are less effective carbon sinks than trees;
   OR
   Disturbances during deforestation releases stored carbon in the soil, leading to further carbon emission into the atmosphere;

2. **Increase in greenhouse gas emission which further traps longwave radiation/ heat, worsening the effects of climate change;**

@1m each

[2]

(iii) Suggest changes Belize can adopt in its agricultural practices to improve agricultural productivity amidst climate change.

1. **Grow drought-tolerant/heat-tolerant crops;**
2. **Improve water irrigation systems;**
3. **Adjustment of crop planting dates to match rainfall patterns;**
4. **Focus the agricultural practices in the northern areas which do not have high incidence of pests;**
5. **Improve pest-management practices so that crops are not easily eaten by pests;**
6. **AVP;**

@1m each, max 2m

[2]

[Total: 10]

For Examiner's Use

Section B

Answer **one** question in this section.

Write your answers on the separate writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

4 **(a)** Explain how the structure of proteins that bind to mRNA is linked to their roles in the regulation of eukaryotic gene expression at post-transcriptional and translational levels. [15]

(b) Explain the significance of bond formation in the process of DNA replication and describe how the end-replication problem arises in eukaryotes. [10]

[Total: 25]

5 **(a)** Describe how eukaryotic ribosomes are formed and explain the role of ribosomes in translation. [15]

(b) It is known that the first eukaryotic cells were unicellular and had evolved from a prokaryotic ancestor.

Explain the functions of telomeres and suggest how telomeres may have originated and evolved in early eukaryotes. [10]

[Total: 25]

[Turn over

For Examiner's Use

4 (a) Explain how the structure of proteins that bind to mRNA is linked to their roles in the regulation of eukaryotic gene expression at post-transcriptional and translational levels. [15]

Structure of RNA binding proteins

1. **Enzyme is specific as the active site has a 3D conformation that is complementary to specific sequences it bind to on mRNA;**
2. **The RNA-binding site on proteins has a 3D conformation which is complementary to specific sequences on mRNA, allowing them to bind;**

Roles at post-transcriptional level (max 8m)

3. **Spliceosome recognises splice sites at introns;**
4. **They excise introns and join together exons in RNA splicing;**
5. **They are involved in alternative splicing resulting in synthesis of more than 1 type of polypeptide from the same pre-mRNA / different combinations of exons;**

6. **Poly(A) polymerase / enzyme recognises the 3' end of pre-mRNA;**
7. **Adds (50-250) adenine nucleotides to it, forming the poly A tail;**
8. **Poly(A) binding proteins recognise the 3' poly A tail;**
9. **Bind to it and slow down the degradation of the 3' end of mRNA by exonucleases / promote export of the mRNA from the nucleus / facilitate the initiation of translation;**

9a. **A longer poly A tail increases the stability of the mRNA;**

10. **Enzymes recognise the 5' end of pre-mRNA;**
11. **and add the modified guanosine cap to it;**
12. **Nuclear export proteins recognise the mature mRNA;**
13. **and help direct it to the nuclear pores for export to the cytoplasm/ slow down the degradation of mRNA by exonucleases/ facilitate the initiation of translation;**

Roles at translational level (max 8m)

14. **Deadenylase / enzyme recognises the mRNA poly (A) tail;**
15. **and shortens it;**
16. **triggering enzymes to remove the 5' modified guanosine cap;**
17. **Once the cap is removed, exonucleases rapidly degrade/ shorten half-life of mRNA;**

18. **Specific proteins recognise and bind to the 3' UTR of the mRNA,**
19. **increasing or decreasing the rate of poly-A tail shortening;**

20. **Translation initiation factors facilitate the binding of the small ribosomal subunit to mRNA / recruit the initiator tRNA, Met-tRNA, to the start codon, AUG;**
21. **When activated or inactivated, translation initiation factors will affect translation of all the mRNAs in a cell;**

22. **Translation repressor proteins recognise and bind to the mRNA at the 5' UTR,**
23. **preventing the small subunit of the ribosome from binding / ribosome assembly;**

24. **Poly(A) polymerase / enzyme (in the cytoplasm) recognises mRNAs with poly A tails of insufficient length;**
25. **adds more adenine nucleotides to it, allowing translation of these mRNAs to begin;**

For Examiner's Use

26. QWC: At least 1 correct point from the structure + 1 correct role in post-transcriptional + 1 correct role in translational;

[Turn over

For Examiner's Use

4

(b) Explain the significance of bond formation in the process of DNA replication and describe how the end-replication problem arises in eukaryotes. [10]

Significance of bond formation (max 7m)

1. **Formation of temporary bonds such as hydrogen bonds, ionic bonds, hydrophobic interactions (any 2) is required for protein/enzyme binding to DNA;**

2. **Binding of helicase to the origin of replication to cause the DNA molecule to unwind and unzip / allow formation of the replication bubble;**
3. **Binding of topoisomerase ahead of the replication fork to make a transient double-stranded break / to prevent the formation of supercoils;**
4. **Binding of single-stranded DNA binding proteins to separated parental DNA strands to prevent reannealing;**
5. **Binding of primase to the template DNA strand to synthesise a short RNA primer;**
6. **Binding of DNA polymerase III to the 3'OH group of an elongating DNA strand for the addition of new deoxyribonucleotides;**
7. **Binding of DNA polymerase I to the RNA primer to remove the primer and replace it with deoxyribonucleotides;**
8. **Binding of DNA ligase to the gap between two Okazaki fragments to form a phosphodiester bond between them;**

MP2 to MP8: max 2m

9. **Formation of hydrogen bonds required for complementary base pairing of nucleotides;**
10. **where adenine pairs with thymine and guanine pairs with cytosine;**
11. **To allow the DNA to be replicated accurately with the correct sequence;**
12. **To allow repair of DNA as proofreading of the sequence is carried out;**
13. **To allow the DNA strands to rewind at the end of DNA replication;**

14. **Formation of phosphodiester bonds between the 3'OH group of one nucleotide and the 5' phosphate of an adjacent nucleotide;**
15. **To allow the formation of a polymer / polynucleotide;**

16. **Hydrophobic interactions between stacked bases;**
17. **To stabilise the DNA structure;**

End replication problem

18. **When the lagging DNA strand is synthesised, many Okazaki fragments are formed, where each fragment has an RNA primer made by primase;**
19. **DNA polymerase I removes the primers and fills the gap with complementary deoxyribonucleotides;**
20. **However, DNA polymerase I cannot fill the gap / there is an absence of an existing 3' OH end at the 5' end of lagging strand / newly synthesised strand / daughter strand;**
21. **Repeated rounds of replication produce shorter and shorter DNA molecules;**

22. **QWC: At least 1m from significance of bond formation + At least 1m from end replication problem;**

For Examiner's Use

5

(a) Describe how eukaryotic ribosomes are formed and explain the role of ribosomes in translation. [15]

Formation of ribosomes

1. **Ribosomes are made up of rRNA and ribosomal proteins;**
2. **rRNA genes/ DNA coding for rRNA undergo transcription in the nucleolus, forming rRNA;**
3. **Genes coding for ribosomal proteins are transcribed in the nucleus to form pre-mRNAs/primary mRNA transcript/mRNA;**
4. **The pre-mRNAs/ primary mRNA transcript undergo post-transcriptional modifications, forming mature mRNAs before they exit the nucleus via nuclear pores;**
5. **They undergo translation in the cytoplasm to form ribosomal proteins;**
6. **Ribosomal proteins are transported from the cytoplasm back into the nucleus;**
7. **rRNA and ribosomal proteins are assembled into the large/ 60S and small/ 40S ribosomal subunits in the nucleolus;**
8. **The ribosomal subunits are then transported to the cytoplasm where they remain separated;**
9. **Assembly of large and small subunits to form 80S ribosomes when translation is initiated;**

Role of ribosomes

10. **Site of polypeptide synthesis/translation;**
11. **Where the mRNA sequence is translated to a sequence of amino acids in a polypeptide;**
12. **Small subunit recognises and binds to the 5' end of the mRNA / near the start codon;**
13. **Large subunit contains the E, P A sites;**
14. **P site – holds the peptidyl-tRNA, which transfers its amino acid(s) to the aminoacyl-tRNA in the A site / the site for the growing polypeptide chain;**
15. **A site – holds the incoming aminoacyl-tRNA, (which accepts the earlier amino acid to form a peptidyl-tRNA);**
16. **A site – also receives the release factor which signals the end of translation;**
17. **Contains peptidyl transferase – the enzyme which catalyses the formation of peptide bonds between adjacent amino acids;**
18. **E site – site where tRNA leaves the ribosome;**
19. **Aligns mRNA in the ribosome, so that anticodons of incoming aminoacyl-tRNA molecules can form hydrogen bonds with mRNA codons via complementary base pairing;**

20. **QWC: At least two correct points for the formation and two correct points for roles of ribosomes;**

[Turn over

For Examiner's Use

5 (b) It is known that the first eukaryotic cells were unicellular and had evolved from a prokaryotic ancestor.

Explain the functions of telomeres and suggest how telomeres may have originated and evolved in early eukaryotes. [10]

Functions of telomeres

1. **Telomeres protect genes at the ends of chromosomes from being eroded during semi-conservative DNA replication due to the end-replication problem;**

2. **Part of the telomeres forms loops at the ends of chromosomes and these loops bind to proteins to protect the ends of chromosomes;**
3. **Telomeres prevent chromosomal end-to-end fusions;**
4. **which can lead to chromosomal aberrations;**
5. **Telomeres protect chromosomal ends from inappropriate degradation by exonucleases;**

6. **Telomere shortening to a critical length also act as signals for cells to enter replicative senescence;**
7. **where the cell cycle is arrested / the cell may undergo apoptosis;**

Evolution of telomeres

8. **There was a change in the structure of DNA molecule from circular to linear form (at one point in the past);**

9. **Variation in presence of non-coding DNA at the ends of DNA molecules occurred in the population of eukaryotic cells due to spontaneous mutation;**
10. **Linear DNA molecules were susceptible to shortening due to the end-replication problem / any other deleterious effect due to lack of telomeres, which acted as a selection pressure;**
11. **Cells with non-coding DNA at the ends of DNA molecules are selected for;**
12. **These cells are more likely to survive and reproduce / have a higher reproductive success, passing on the presence of the telomere to daughter cells;**
13. **Over time, there is an increase in frequency of *cells* having telomeres in the gene pool of the cell population;**

14. **QWC: At least 2m for the function of telomeres + 1m for the evolution of telomeres;**